ARGUMAT



ZEPHYR

Ambient Air Bread Cooler
Combined Ambient & Cold Air Bread Cooler

Contents

Introduction

- Bread cooling
- Operating principle
- Range & dimensions
- Components
- Product definition
- Recommendations

Arguments

- Optimisation of bread cooling time
- Stabilisation of the T° in the bakery
- Fewer trolleys & baking trays needed
- Optimisation of cooling before blast-freezing
- Improved slicing rate
- Preservation of product quality after bagging

The Pro’s

Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

BREAD COOLING



BREAD COOLING

Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

➢ DEFINITION

- **Bread cooling** is the period that starts immediately after the baking cycle
- During this phase part of the water contained in the product tuns into steam
- The objective? To bring the product to room temperature fastly

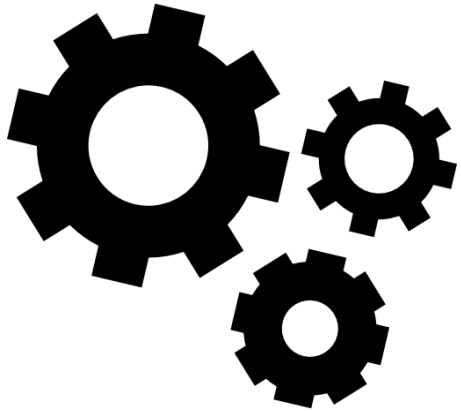
Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

OPERATING PRINCIPLE



Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

OPERATING PRINCIPLE

➢AMBIENT AIR BREAD COOLER

- A bread cooling compartment, by ambient air ventilation, for 2 or 4 trolleys
- Trolleys placed in the bread cooling compartment as they exit the oven
- Air is blown through the trolleys thanks to the blower slots positioned over the entire height of the inner side panel of the compartment
- One or several fans located on top on the opposite side collect the hot air loaded with moisture.





ONLY AVAILABLE ON 2 AND 4 TROLLEYS VERSION

Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

OPERATING PRINCIPLE

➢COMBINED AMBIENT & COLD AIR BREAD COOLER

- A compartment separated in two sections :
  - The 1st one for ambient air cooling (operation described on the previous slide)
  - The second one dedicated to cooling by refrigerated air

- The sections are separated by a PVC strip curtain

ONLY AVAILABLE ON 2 TROLLEYS VERSION

Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

OPERATING PRINCIPLE

➢COMBINED AMBIENT & COLD AIR BREAD COOLER

- Once the ventilation phase is over, the trolleys are placed in the refrigerated section

- Reduction of the product temperature thanks to the evaporators connected to the fans



ONLY AVAILABLE ON 2 TROLLEYS VERSION

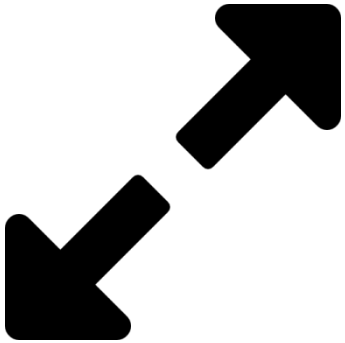
Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

RANGE & DIMENSIONS



RANGE & DIMENSIONS

Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

Cooling technology	***Number of 600x800 trolleys***		
	2	***4***	***4***
	By ambient air		***By combination (ambien and cold air)***
Extractors			
Number	2	3	2
Diameter (Ø)	200	200	200
Dimensions			
Overall height	2560	2560	2670
Chamber height	2090	2090	2090
Overall width	1550	1550	1550
Depth on the floor	1750	3350	3350
Depth with both doors open	1870	3470	3470
Doors opening depth	3990	5590	5590
Cooling features			
Fluid	-	-	R448A
Power	-	-	3810W at -10/+32°C

Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

COMPONENTS



Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

COMPONENTS

➢ COOLER LAYOUT

- PVC strip curtain for air flows separation (only for cold air coolers)



Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

COMPONENTS

➢ COOLER LAYOUT

- Stainless steel interior to ensure perfect hygiene and ease of cleaning
- Inner rails to preserve the panels from possible shocks due to the flow of trolleys



Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

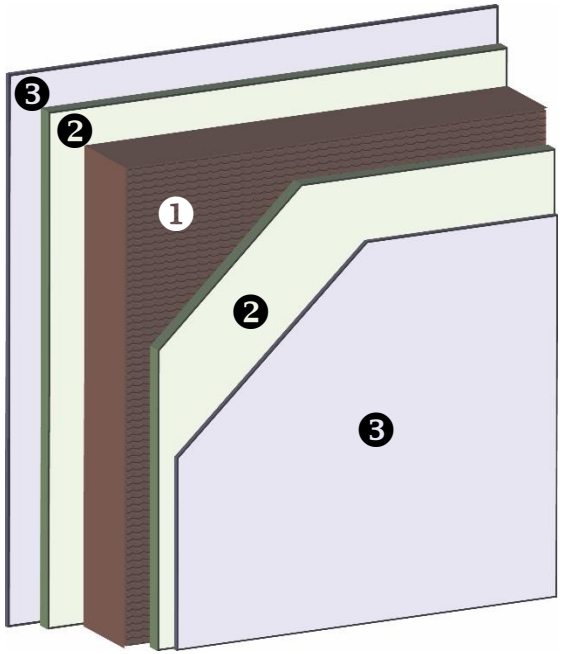
COMPONENTS

➢ PANELS: 60 mm THICKNESS

❶ Polyrethane foam: 45kg/m^3

❷ **Steel** cladding

❸ **120µ PVC cling** film



COMPONENTS

Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

➢ CONTROL

ON/OFF + temperature display in case of refrigerated section inside compartment



Punch Push Button: starting a cycle

Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

PRODUCT FILE



PRODUCT FILE

Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

Bread cooler		
Capacity		
Number of 600x800 trolleys	2	4
Control		
Electromechanical	■	■
Finishing of panels and doors		
Stainless steel interior/PVC coated steel exterior	■	■
Cooling technology		
By ambient air	■	■
By combination (ambient and cold air)	❖	€
Refrigeration units		
Standard condensing unit	❖	■
« Silensys » low noise level condensing unit	❖	€
Connected to central unit	❖	Consult us
Miscellaneous		
Stainless steel protection rail	■	■
PVC slat curtain between compartments (only for version with cold air unit)	❖	■
Connection		
~230V 50-60Hz (single phase) for ambient air cooler only	■	■
3N~400V 50Hz (three phase) for ambient air +cold air cooler	❖	■

Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

RECOMMENDATIONS



Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

RECOMMENDATIONS

➢ AMBIENT AIR BREAD COOLER OR COMBINED AMBIENT & COLD AIR BREAD COOLER ?

- Cold air coolers are recommended in the following cases:
  - Hot climates
  - Humid climates





Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

RECOMMENDATIONS

➢ NUMBER OF COOLERS PER OVEN

- For rotary ovens







4-TROLLEY COOLER FOR TWO ROTARY OVENS

Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

.

ARGUMENT



Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

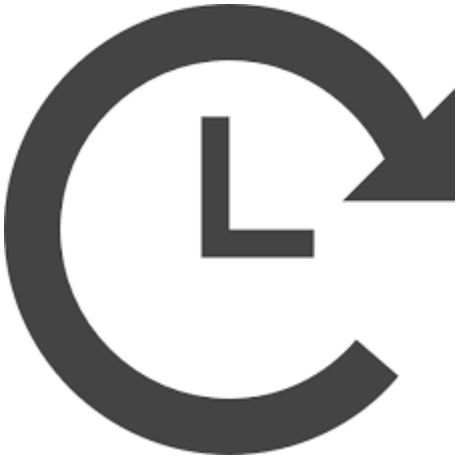
Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

ARGUMENT

➢ OPTIMISATION OF BREAD COOLING TIME

- This equipment allows faster cooling down of products
- Bread cooling time in open air is superior than bread cooling time in the cooler



Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

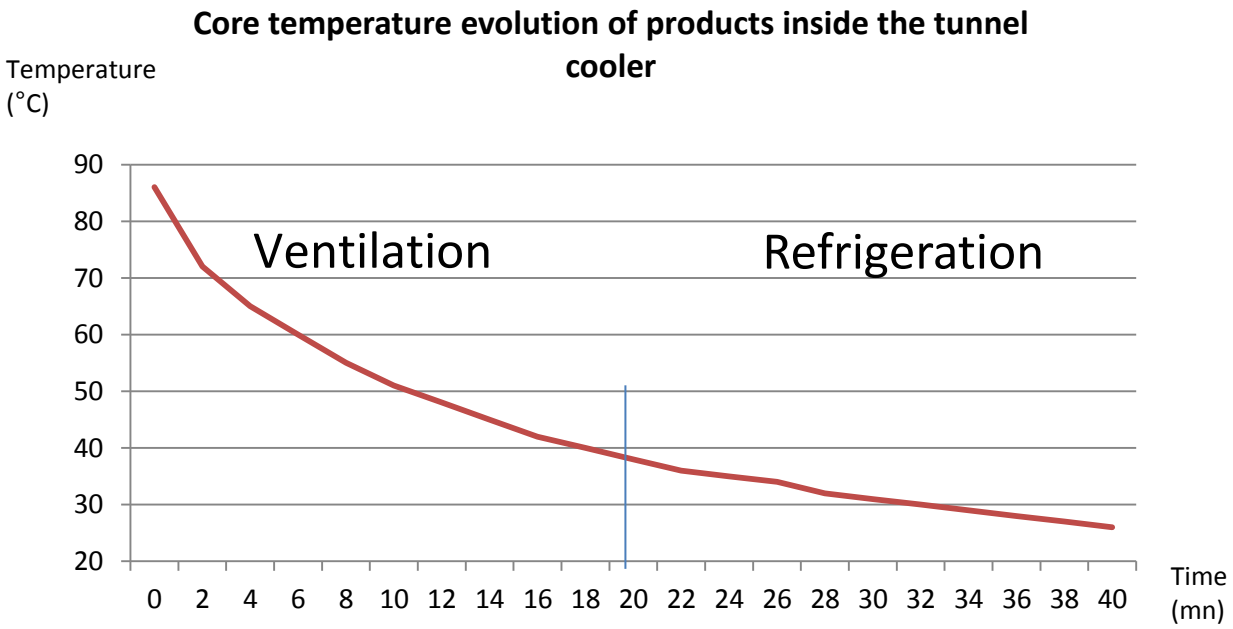
The Pro's

ARGUMENT

➢ OPTIMISATION OF BREAD COOLING TIME

Combined ambient & cold air bread cooler

- "Ambient air" cooling down time inside a bakery at +20°C : 1h30/2h00
- Cooling down time inside a « cold air » bread cooler with a refrigeration unit : 40mn to reach 26°C at core (test based on baguettes)



Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

ARGUMENT

➢ LIMITATION OF TEMPERATURE RISE IN THE BAKERY

- The trolleys give off heat as soon as they leave the oven
- Placing the products in a bread cooler limits temperature rise inside the bakery
- Dough mixing quality depends on the room temperature
- Bread cooler = controlled bakery temperature

Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

ARGUMENT

➢ FEWER TROLLEYS & FEWER BAKING TRAYS NEEDED

- As bread cooling time gets shorter, fewer trolleys are needed (faster trolleys rotation)
- Trolleys are stored in the cooler hence do not impede the traffic in the bakery

ARGUMENT

Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

➢ OPTIMISATION OF THE COOLING BEFORE BLAST-FREEZING

- After baking, the trolleys can be placed in the bread cooler, allowing the products to reach an optimal temperature before blast-freezing (pre-baked products)

- Improves the consistency of finished products



Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

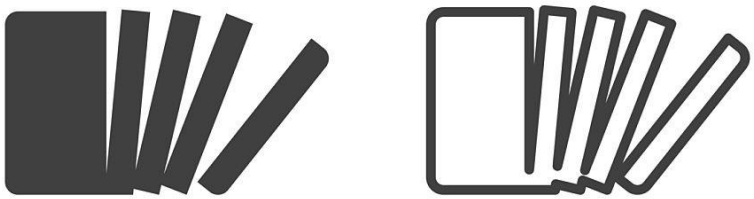
Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

ARGUMENT

➢ IMPROVES THE SLICING AND PACKAGING RATE

- Hot products are difficult to slide and will soften if bagged too fast
- On the contrary: cooled down products retain their crispiness and can be immediately bagged, sliced or unsliced



Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

ARGUMENT

➢ PRESERVES PRODUCT QUALITY AFTER BAGGING

- If bread is bagged when still hot, it will produce condensation in its container and the product will become soften
- The bread cooler is **THE** solution to allow quick bagging while maintaining the crispiness of our products.





Contents

Introduction
Bread cooling
Operating principle
Range & dimensions
Components
Product file
Recommendation

Arguments
Optimisation of bread cooling time
Temperature stabilisation
Fewer trolleys needed
Before deep-freezing
Slicing rate
Bagging

The Pro's

.

THE PRO'S



THE PRO'S

CUSTOMER GROUP	SUPERMARKET	SEMI-INDUSTRIAL ARTISANS
Optimisation of bread cooling time	VV	VV
Stabilisation of the T° inside the bakery	V	V
Fewer trolleys & baking trays needed	V	V
Optimisation of cooling before blast-freezing	V	VV
Slicing rate	VV	VV
Product quality after bagging	VV	VV